

Finding the Joy: Investigating reward-processing during word learning in children with and without dyslexia

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1. INTRODUCTION

- While reading, we extract the meaning of new words from context, honing our vocabulary in the process.
- Adults find learning new words from context enjoyable, showing increased activity in reward processing regions such as the ventral striatum.
- Coupling between the ventral striatum and language learning areas fuels human motivation to learn language.

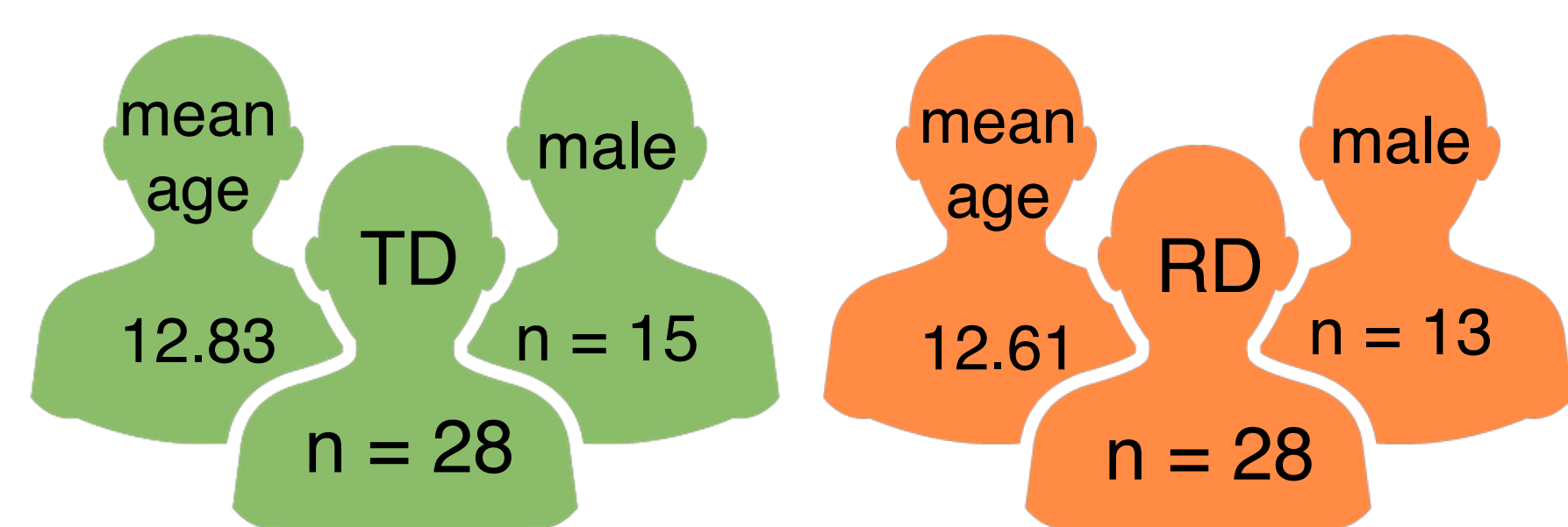
2. RESEARCH QUESTIONS

- Do children also use the same reward processing systems during word learning?
- Does this activity differ in children with poor reading ability?

3. GROUP DEFINITION

Group	Nonverbal IQ > 70	1 ≥ SD below mean on reading tests	History of reading difficulties
TD	✓		
RD	✓	✓	✓

4. PARTICIPANTS



11-13 year-olds, grown up in the UK speaking English

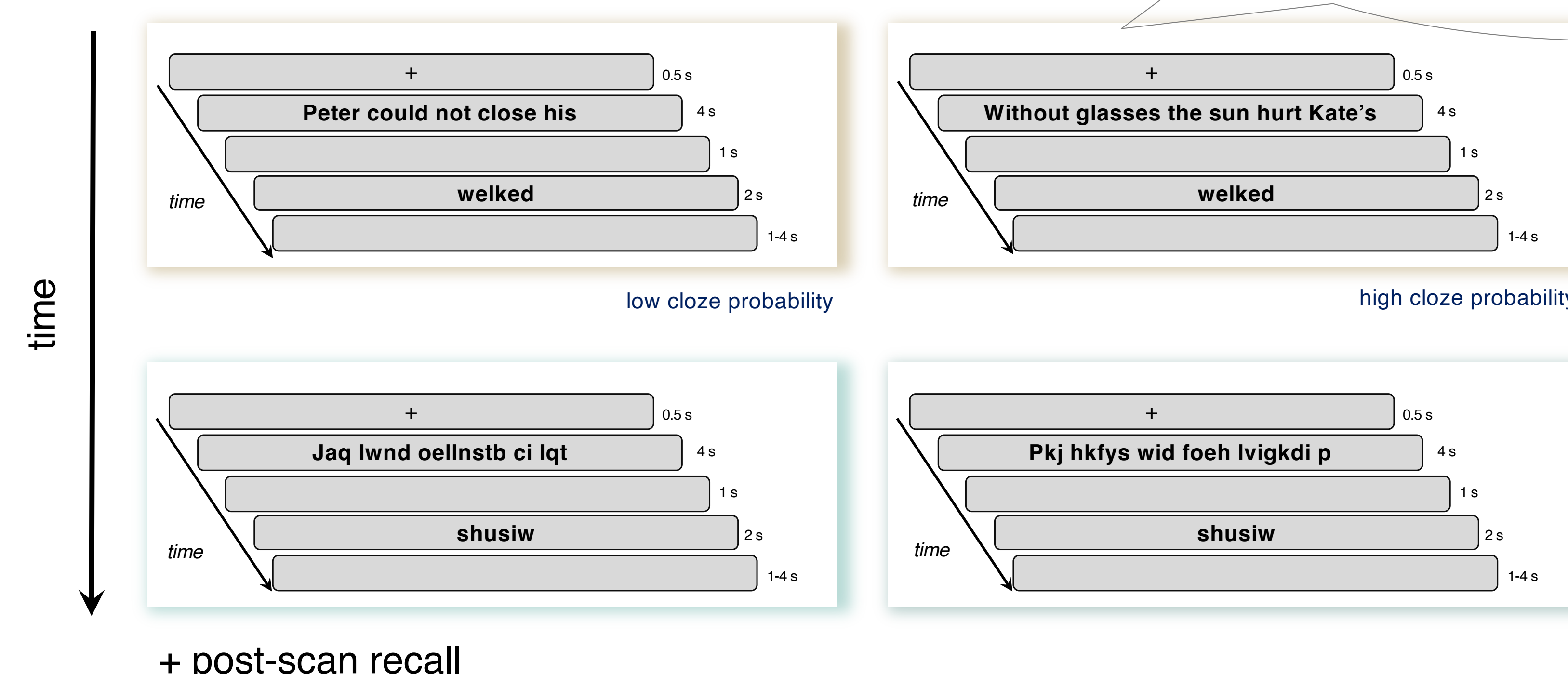
Behavioural measures

- Receptive vocabulary; expressive grammar skills; non-verbal intelligence; word & non-word reading, reading comprehension skills

5. IN SCANNER TASKS

A. Contextual word learning paradigm

- 36 pseudo-words in 6 blocks, block 3 sets
- Each set 2 pseudo-word sentence pairs + 1 non-readable sentence pair

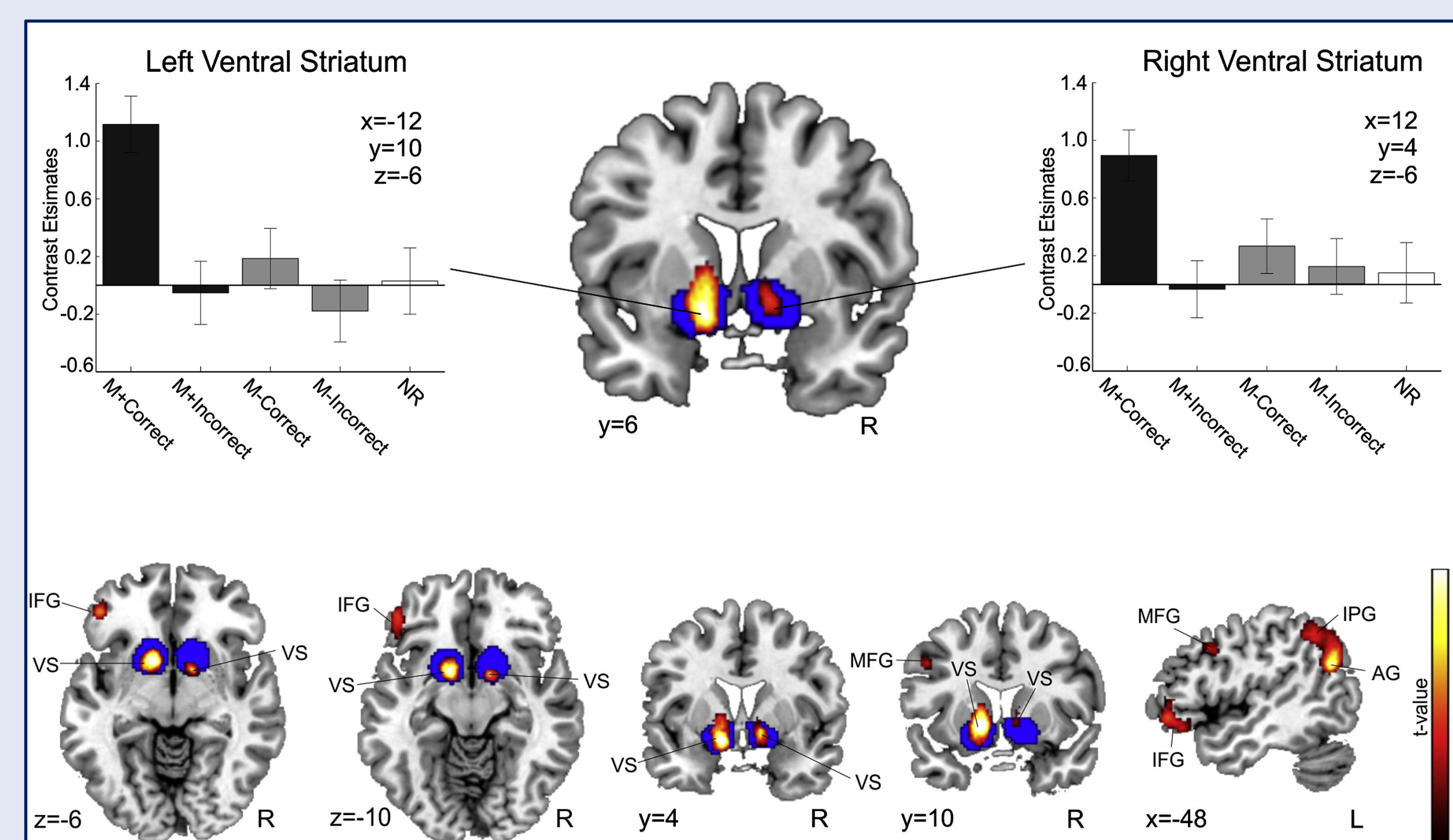
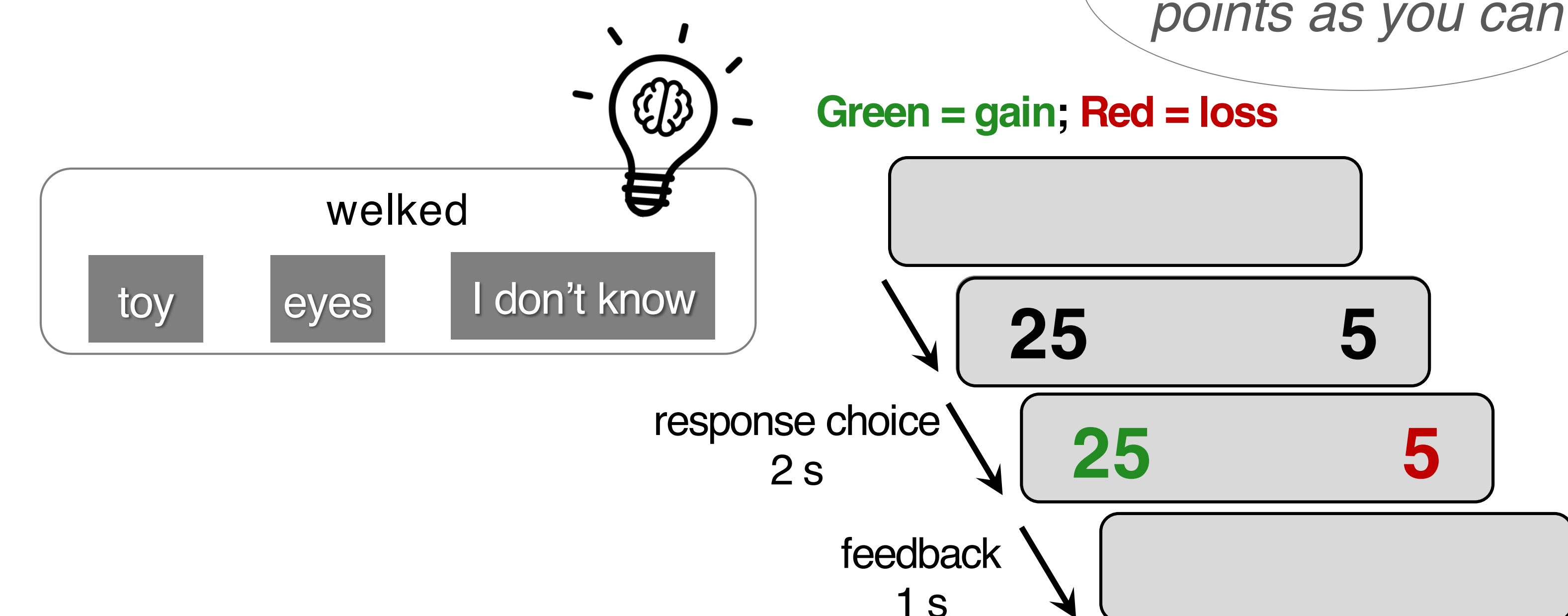


"try to learn what the meaning of the made-up words might be by reading the sentences..."

B. Reward localiser

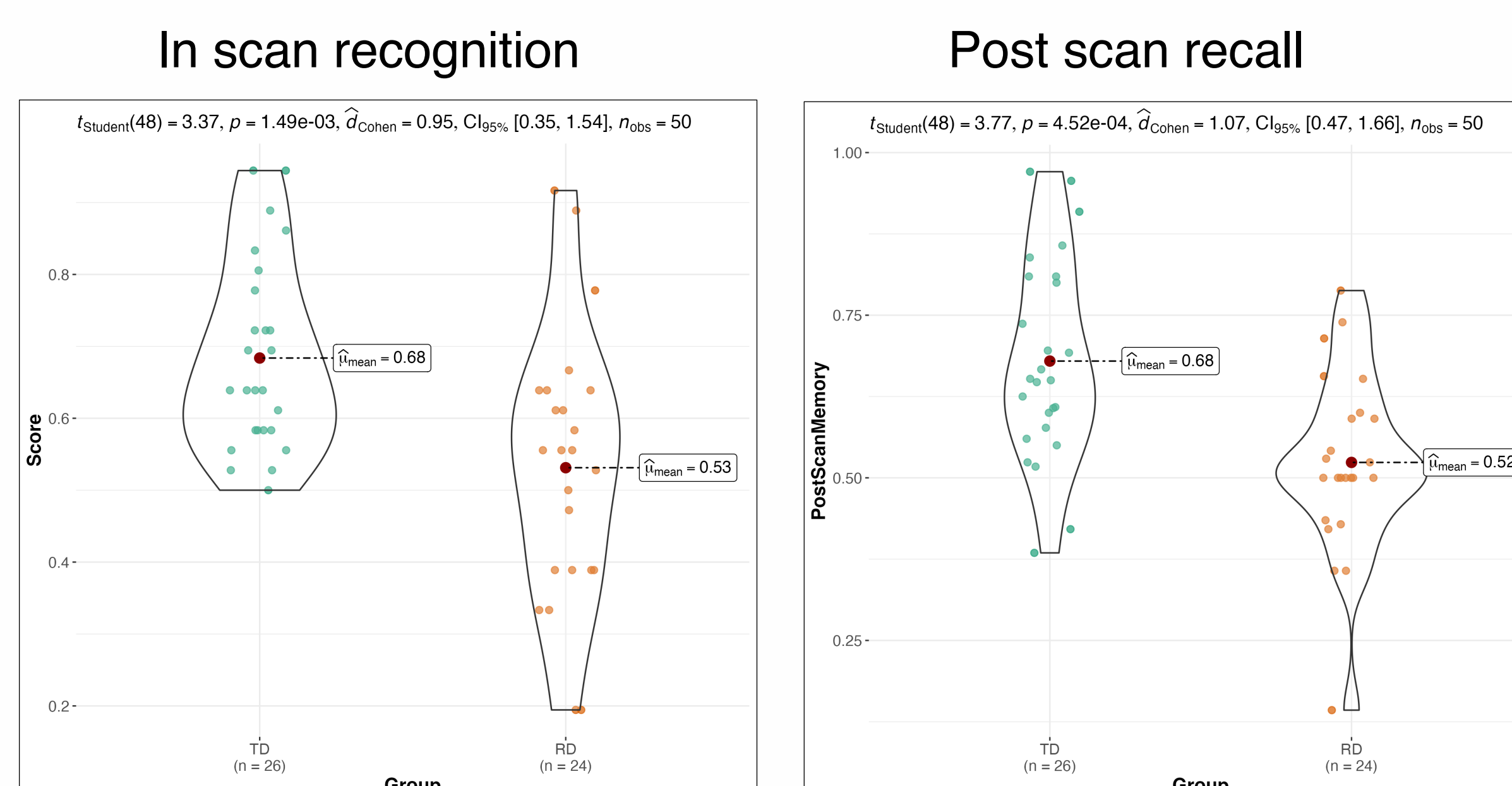
- 30 win + 30 loss trials
- 15 'boost win' + 15 boost loss' trials

"try to earn as many points as you can"

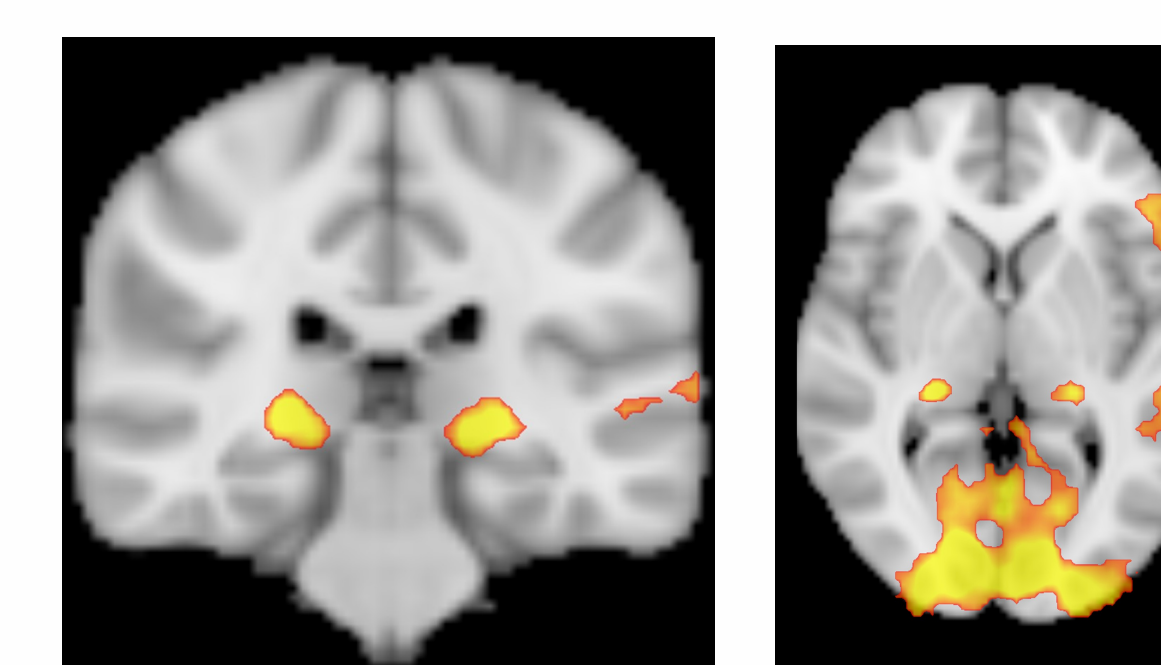


Ripollés et al. 2014 | Current Biology

6.a PRELIMINARY BEHAVIOURAL RESULTS



6.b PRELIMINARY fMRI RESULTS



- Group average activity for reading sentences > unreadable sentences (TD = 13; RD = 17), displayed at a whole brain threshold of $Z > 4.42$

7. PLANNED ANALYSES

- Investigate whether children show increased BOLD activity in subcortical reward circuits, and increased connectivity between the ventral striatum and language regions, when successfully learning the meaning of new words.
- Assess whether the neural response associated with successful word learning is different in children with dyslexia.
- Take a continuum approach, assessing if reading ability correlates with enhanced neural activity in A) reading-relevant cortical regions and B) subcortical reward-processing circuitry.

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